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Cleaning device with fluid tank empty detection

Abstract

Various cleaning devices with fluid tank empty detection and methods related thereto are provided. In general, a cleaning device can be configured to determine whether a fluid container is empty using an electromotive force (EMF) at a fluid pump configured to pump fluid from the fluid container. When the fluid container becomes empty, fluid cannot be supplied from the cleaning device to a surface until more fluid is added to the fluid container. Thus, by determining whether the fluid container is empty, a user of the cleaning device may be made aware when the fluid container becomes empty and needs refilling (or replacement).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of International Patent Application No. PCT/CN 2023/110777 entitled "Cleaning Device With Fluid Tank Empty Detection" filed Aug. 2, 2023, which is hereby incorporated by reference in its entirety.

FIELD

(1) The present disclosure generally relates to cleaning devices with fluid tank empty detection.

BACKGROUND

(2) Conventional cleaning devices, such as dry vacuums and wet vacuums, perform cleaning operations using suction to take in waste. Dry vacuums operate through the use of suction and may employ a brushroll or other agitator to assist in freeing the waste from a surface. Wet vacuums operate through the use of suction and an agitator or pad, but they also supply fluid to the to-be-cleaned surface in order to assist in removal of waste. The supply of fluid can occur directly, with fluid being sprayed onto a surface, or indirectly, with fluid being sprayed onto an applicator such as an agitator. Some vacuums are either a dry vacuum or a wet vacuum. Some vacuums are wet dry vacuums able to provide the functionality of a dry vacuum and a wet vacuum.

(3) For wet vacuums and wet dry vacuums, the fluid to be supplied to a surface can be stored on board the device in a fluid container such as a fluid supply tank. The amount of fluid in the fluid container decreases as fluid is supplied to the surface to be cleaned. When the fluid container becomes empty, fluid cannot be supplied from the cleaning device to a surface until more fluid is added to the fluid container. However, a user of the cleaning device may not be aware when the fluid container becomes empty and needs refilling (or replacement) because a user may forget to check the fluid container's fill level before attempting to supply fluid from the cleaning device, the container may be opaque and thus have its fill level obscured from user view; the fluid container may be disposed within another component (e.g., a housing or other component) of the cleaning device and thus have its fill level obscured from user view, or another reason. User experience may thus be degraded by fluid not being available in the fluid container when the user desires to clean using the cleaning device.

(4) Accordingly, there remains a need for improved devices, systems, and methods for cleaning devices.

SUMMARY

(5) In general, systems, devices, and methods for cleaning devices with fluid tank empty detection are provided.

(6) In one aspect, a system is provided that in one embodiment includes a fluid pump of a cleaning device and includes a controller. The fluid pump is configured to pump a fluid from a fluid supply tank for delivery to a surface to be cleaned by the cleaning device. The fluid pump includes a motor configured to drive the pumping of the fluid of and to generate an electromotive force (EMF). The controller is configured to receive from the fluid pump a signal indicative of the EMF and is configured to determine, based on the received signal, whether the fluid supply tank is substantially empty.

(7) The system can have any number of variations. For example, the controller can be configured to, in response to determining that the fluid supply tank is substantially empty, cause a user notification to be provided via the cleaning device indicating that the fluid supply tank is substantially empty.

(8) For another example, the controller receiving the signal can include the controller receiving a plurality of signals from the fluid pump, each of the plurality of signals can be indicative of the EMF generated by the motor during a period of time, and the controller determining whether the fluid supply tank is substantially empty can include the controller comparing the plurality of signals with a predetermined threshold EMF value. Further, the predetermined threshold EMF value can be a preset value that does not change: or the controller can be configured to determine whether the plurality of signals are outside of a predetermined range, if the plurality of signals are not outside of the predetermined range, the predetermined threshold EMF value can remain the same, and if the plurality of signals are outside of the predetermined range, the controller can be configured to change the predetermined threshold EMF value to a new predetermined threshold EMF value. Further, the controller can be configured to calculate the new predetermined threshold EMF value based on the plurality of signals.

(9) For yet another example, the motor can include a rotor and a stator.

(10) For still another example, the system can further include the fluid supply tank.

(11) In another embodiment, a system includes a processor and a memory storing instructions that, when executed by the processor, cause the processor to perform operations including causing a fluid pump of a cleaning device to pump a fluid from a fluid supply tank for delivery to a surface to be cleaned by the cleaning device. The fluid pump includes a motor configured to drive the pumping of the fluid of and to generate an EMF. The operations also include determining, based on generated EMF, whether the fluid supply tank is substantially empty.

(12) The system can vary in any number of ways. For example, the operations can further include, in response to determining that the fluid supply tank is substantially empty, causing a user notification to be provided via the cleaning device indicating that the fluid supply tank is substantially empty.

(13) For another example, determining whether the fluid supply tank is substantially empty can include comparing EMF data received by the processor from the fluid pump with a predetermined threshold EMF value. Further, the predetermined threshold EMF value can be a preset value that does not change: or the operations can further include determining whether the EMF data are outside of a predetermined range, if the EMF data is not outside of the predetermined range, the predetermined threshold EMF value can remain the same, and if the EMF data is outside of the predetermined range, the operations can further include changing the predetermined threshold EMF value to a new predetermined threshold EMF value. Further, the operations can further include calculating the new predetermined threshold EMF value based on the EMF data.

(14) For yet another example, the motor can include a rotor and a stator.

(15) For still another example, the system can further include the fluid supply tank.

(16) In another aspect, a method is provided that in one embodiment includes causing, using a controller, a fluid pump of a cleaning device to pump a fluid from a fluid supply tank for delivery

to a surface to be cleaned by the cleaning device. The fluid pump includes a motor configured to drive the pumping of the fluid of and to generate an EMF. The method also includes determining, using the controller and based on generated EMF, whether the fluid supply tank is substantially empty.

(17) The method can have any number of variations. For example, the method can further include, using the controller and in response to determining that the fluid supply tank is substantially empty, causing a user notification to be provided via the cleaning device indicating that the fluid supply tank is substantially empty.

(18) For another example, determining whether the fluid supply tank is substantially empty can include comparing, using the controller, EMF data received by the processor from the fluid pump with a predetermined threshold EMF value. Further, the predetermined threshold EMF value can be a preset value that does not change; or the method can further include determining, using the controller, whether the EMF data are outside of a predetermined range, if the EMF data is not outside of the predetermined range, the predetermined threshold EMF value can remain the same, and if the EMF data is outside of the predetermined range, the method can further include changing, using the controller, the predetermined threshold EMF value to a new predetermined threshold EMF value. Further, the method can further include calculating, using the controller, the new predetermined threshold EMF value based on the EMF data.

(19) For yet another example, the motor can include a rotor and a stator.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) This disclosure will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a perspective view of one embodiment of a cleaning device;

(3) FIG. 2 is a front view of the cleaning device of FIG. 1;

(4) FIG. 3 is a side view of the cleaning device of FIG. 1;

(5) FIG. 4 is a side cross-sectional view of the cleaning device of FIG. 1;

(6) FIG. 5 is a perspective view of a body assembly of the cleaning device of FIG. 1 with one embodiment of a fluid supply tank and one embodiment of a recovery tank removably coupled with the body assembly;

(7) FIG. 6 is a side view of the body assembly, the fluid supply tank, and the recovery tank of FIG. 5;

(8) FIG. 7 is a perspective view of the body assembly of FIG. 5 without a fluid supply tank and a recovery tank removably coupled with the body assembly;

(9) FIG. 8 is a side view of the body assembly of FIG. 7;

(10) FIG. 9 is a perspective view of the fluid supply tank of FIG. 5;

(11) FIG. 10 is a side view of the fluid supply tank of FIG. 9;

(12) FIG. 11 is a side cross-sectional view of the fluid supply tank of FIG. 9;

(13) FIG. 12 is an exploded view of the fluid supply tank of FIG. 9;

(14) FIG. 13 is a top view of a head assembly of the cleaning device of FIG. 1, the head assembly having no upper housing and showing components used in wet vacuum modes;

(15) FIG. 14 is a perspective view of a portion of a handle assembly of the cleaning device of FIG. 1;

(16) FIG. 15 is a perspective view of a partial portion of the handle assembly of FIG. 14;

(17) FIG. 16 is a perspective view of another partial portion of the handle assembly of FIG. 14;

(18) FIG. 17 is a perspective view of the handle assembly of FIG. 14;

(19) FIG. 18 is a schematic view of a portion of the cleaning device of FIG. 1;

- (20) FIG. 19 is one embodiment of a circuit diagram for the cleaning device of FIG. 1;
- (21) FIG. 20 is a graph showing EMF over time;
- (22) FIG. 21 is another graph showing EMF over time;
- (23) FIG. 22 is yet another graph showing EMF over time;
- (24) FIG. 23 is still another graph showing EMF over time;
- (25) FIG. 24 is another graph showing EMF over time;
- (26) FIG. 25 are other graphs showing EMF over time;
- (27) FIG. 26 is a flowchart of one embodiment of a method; and
- (28) FIG. 27 is a flowchart of another embodiment of a method.

DETAILED DESCRIPTION

(29) Certain embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices, systems, and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices, systems, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

(30) Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

(31) Various cleaning devices with fluid tank empty detection and methods related thereto are provided. In general, a cleaning device can be configured to determine whether a fluid container is empty using characterizing an electromotive force (EMF) at a fluid pump configured to pump fluid from the fluid container. When the fluid container becomes empty, fluid cannot be supplied from the cleaning device to a surface until more fluid is added to the fluid container. Thus, by determining whether the fluid container is empty, a user of the cleaning device may be made aware when the fluid container becomes empty and needs refilling (or replacement). User experience may thus be improved by allowing fluid to be available in the fluid container when the user desires to clean using the cleaning device.

(32) Some traditional cleaning devices can include a sensor, e.g., a photoelectric sensor, configured to detect whether a fluid container is empty. However, such sensors are expensive and add cost to the cleaning device. The fluid tank empty detection described herein does not use a photoelectric sensor or other type of sensor in detecting whether a fluid container is empty and thus avoids cost of such a sensor. Additionally, such sensors are susceptible to damage by being exposed to fluid since the sensors are used in association with a fluid container. Whether or not the fluid container is empty cannot be determined if the sensor is damaged, thereby reducing functionality of the cleaning device and reducing user experience. The fluid tank empty detection described herein does not use a photoelectric sensor or other type of sensor in detecting whether a fluid container is empty and thus avoids reducing functionality of the cleaning device and reducing user experience due to a damaged sensor. Further, the fluid tank empty detection described herein uses existing components of a cleaning device, e.g., a fluid pump and a controller, and therefore does not add any additional components to the cleaning device in order to achieve fluid tank empty detection, which may help maintain a lower cost of the cleaning device, may free real estate for other components, and/or may allow for a smaller cleaning device.

(33) The systems, devices, and methods described herein are not limited to cleaning devices. The systems, devices, and methods described herein can be similarly used with other types of devices having a fluid supply tank configured to be refilled (or replaced).

(34) Various embodiments of cleaning devices are described, for example, in U.S. Pat. No. 11,484,172 entitled “Wet Dry Appliance” issued Nov. 1, 2022, U.S. patent application Ser. No. 17/950,942 entitled “Wet Dry Appliance” filed Sep. 22, 2022, and U.S. patent application Ser. No. 17/653,558 entitled “Low Cost Cleaning Head For Cleaning Devices” filed Mar. 4, 2022, which are hereby incorporated by reference in their entireties.

(35) FIGS. 1-4 illustrate one exemplary embodiment of a cleaning device **10**. The illustrated cleaning device **10** includes a head assembly **100**, a body assembly **200**, a handle assembly **300**, and a vacuum assembly (obscured in FIGS. 1-4). The cleaning device **10** is shown disposed atop a charging mat **400**, but the cleaning device **10** can be configured to not be usable with a charging mat. The device **10** also includes fluid delivery and fluid recovery assemblies. In this illustrated embodiment, the handle assembly **300** includes a handle **310** and a stem **320**, and the body assembly **200** includes a body housing **210** coupled to the stem **320**. The head assembly **100** is coupled to the body housing **210** opposite the stem **320**. The head assembly **100** includes a head housing **110** and includes small wheels (obscured in FIGS. 1-4) and large wheels **112L**, **112R** rotatably coupled to the head housing **110** and configured to allow the cleaning device **10** to roll along a surface. The head assembly **100** also includes a brushroll (obscured in FIGS. 1-4) disposed in the head assembly **100** and configured to rotate during operation of the cleaning device **10**. The cleaning device **10** in this illustrated embodiment is described as including a brushroll, but the cleaning device **10** can include another type of agitator.

(36) The vacuum assembly is disposed within the head and body assemblies **100**, **200** and is capable of taking in fluid, dirt, debris, and other waste through suction and storing it within the cleaning device **10**. As in this illustrated embodiment, the vacuum assembly can include a motor and a motor fan. The motor and motor fan can be entirely contained in a motor housing disposed within the body assembly **200**. Hosing **230** is coupled to the motor fan and runs through the body assembly **200** to the head assembly **100** to allow the motor to generate a suction force to draw waste into the device **10**. Waste taken in by the vacuum assembly through the hosing **230** is deposited into a recovery tank **500** removably disposed within the body assembly **200**. In other embodiments, the recovery tank **500** may not be removable.

(37) The cleaning device **10** also include a fluid supply tank **600**, also referred to herein as a “fluid container.” The fluid supply tank **600** is configured to contain fluid therein that is configured to be supplied from the device **10** to a floor or other area to be cleaned. The fluid contained in the fluid supply tank **600** is clean fluid. The clean fluid can be water, a cleaning solution, or other fluid configured to aid in cleaning the floor or other area to which the fluid is delivered from the device **10**. Once delivered from the device **10**, the fluid can be mixed with waste (e.g., dirt, dirty fluid, debris, etc.). The cleaning device **10** is configured to draw the waste and fluid mixed therewith into the device **10** with suction generated by the motor and to deposit the drawn-in material in the recovery tank **500**.

(38) As previously indicated, the cleaning device **10** is configured to operate in a wet cleaning mode and a dry cleaning mode. The device **10** can operate in only one of the wet and dry cleaning modes at a time or can operate in the wet and dry cleaning modes simultaneously. The cleaning device **10** in this illustrated embodiment is configured operate normally in a dry cleaning mode in which the vacuum assembly is employed to suction in waste, but upon selection of the wet cleaning mode, the cleaning device **10** will also begin to emit fluid to aid in the cleaning process. As discussed further below, the device **10** includes a user interface **302** (see FIG. 2) configured to allow user selection of cleaning mode.

(39) Dry cleaning modes generally include modes related to traditional vacuuming operations, such as vacuuming on hard surfaces or on softer surfaces, such as carpet. Dry cleaning modes rely on

suction to take waste into the recovery tank of the cleaning device **10** for convenient disposal. In some dry cleaning modes, the brushroll can rotate to agitate waste on a cleaning surface. The brushroll can loosen the waste while simultaneously directing it toward a suction intake of a cleaning device. In other dry cleaning modes, the brushroll does not rotate (or, in other embodiments, a brushroll is not present), and instead, suction is relied on alone to force waste into the cleaning device **10**.

(40) Wet cleaning modes generally include the cleaning device **10** supplying fluid from the fluid supply tank **600** either directly or indirectly to a surface to aid in cleaning. The supplied fluid can act to loosen waste stuck to the surface, and the dirtied fluid can be taken into the cleaning device through suction or other means. In some wet cleaning modes, like some dry cleaning modes described above, the brushroll can further assist in loosening waste off the surface and directing it toward a suction intake. In these wet cleaning modes, the fluid can be supplied directly to the brushroll in order to simultaneously apply the fluid to the surface while agitating the waste found on the surface. In other wet cleaning modes, the fluid can be supplied directly to the surface and the brushroll can agitate the wetted surface. In still other modes, the fluid can be supplied directly to the surface and the brushroll can remain stationary (or, in other embodiments, a brushroll is not present), thereby cleaning the surface with fluid and suction alone.

(41) As shown in FIGS. **5** and **6**, the body assembly **200** is configured to be operatively coupled to the head assembly **100** via an articulator **250**. The articulator **250** is coupled to the bottom of the body assembly **200** and is configured to at least partially disposed within the head assembly **100**. The illustrated articulator **250** is configured to articulate about two degrees of freedom. A first point of articulation **254**, allowing for articulation about a first degree of freedom, is mounted within the head assembly **100**. The first point of articulation **254** allows for the body assembly **200** to pivot between a forward direction and a backward direction, as indicated by the arrows A-A in FIGS. **5** and **6**. A second point of articulation **256**, located above the first point of articulation **254**, allows for the body assembly **200** to pivot between a left direction and a right direction, as indicated by the arrows B-B in FIG. **5**. One or both of the first and second points of articulation **254**, **256** can be articulated at a given time. Further, in other embodiments, the body assembly **200** can be configured to articulate in any number of degrees of freedom about any number of points of articulation.

(42) As also shown in FIGS. **5** and **6**, the body housing **210** of the body assembly **200** includes a housing base **210a**, a front side **210b**, a rear side **210c** extending upward from the housing base **210a**, and a top side **210d**. The top side **210d** of the body housing **210** is coupled to the handle assembly **300**, which extends from the body assembly **200** in a direction opposite the head assembly **100**.

(43) As shown in FIGS. **7** and **8**, the body housing **210** of the body assembly **200** includes first and second cavities **210e**, **210f** configured to removably receive components of the cleaning device **10**. The first cavity **210e** is configured to removably receive the recovery tank therein, as shown in FIGS. **5** and **6**. The second cavity **210f** is configured to removably seat the fluid supply tank **600** therein, as shown in FIGS. **5** and **6**. FIGS. **7** and **8** show the body assembly **200** with the recovery tank and the fluid supply tank **600** removed from the first and second cavities **210e**, **210f**, respectively.

(44) The first cavity **210e**, located in the front side **210b** of the body housing **210**, is sized to removably receive the recovery tank such that, when retained in the first cavity **210e**, the recovery tank occupies the entirety of a lower region of the front side **210b** of the body housing **210**. The recovery tank is removable from the body housing **210** after actuation of a latch assembly **430** (see FIGS. **5** and **6** extending outward from an upper extent of the recovery tank. The actuation of the latch assembly **430** releases the recovery tank from engagement with a retaining slot **214** (see FIG. **8**) that is located toward the front of the first cavity **210e**.

(45) The second cavity **210f** is located in an upper front portion **210b** of the body housing **210** and

occupies a substantial portion of the top side **210d**. The second cavity **210gf** is sized to removably receive the fluid supply tank **600**. A fluid tank switch **212** (see FIGS. **7** and **8**) is disposed in the top side **210d** of the body housing **210**. With the fluid supply tank **600** seated in the second cavity **210f**, a tank engagement feature **216** (see FIG. **7**) of the body assembly **200** extends from the body housing **210** and is engaged with the fluid supply tank **600** to lock the fluid supply tank **600** to the body housing **210**. When the fluid tank switch **212** is actuated, the tank engagement feature **216** recedes into the body housing **210**, which allows the fluid supply tank **600** to be removed from the second cavity **210f**.

(46) FIGS. **9-12** illustrate the fluid supply tank **600** as a standalone element. As shown in FIGS. **9-12**, the fluid supply tank **600** includes a valve cap **612** removably threaded to a fluid tank **614**. The fluid tank **614** is divided into an upper tier **614a** and a lower tier **614b**. Each of the tiers **614a**, **614b** has a substantially hemi-cylindrical shape in this illustrated embodiment but may have other shapes. The upper tier **614a** is shaped to conform with the overall form of the body housing **210** and provides an outer limit for the upper front face **210b** of the body housing **210**. The lower tier **614b** is smaller than the upper tier **614a** in this illustrated embodiment and is configured to be received internally within the body housing **210**. The fluid tank **614** defines a hollow interior, which receives the fluid to be supplied by the cleaning device **10** during a wet cleaning operation.

(47) The valve cap **612** of the fluid supply tank **600** permits one-way flow of fluid therethrough from the hollow interior of the fluid tank **614** to externally thereof. The valve cap **612** is configured to be received in the second cavity **210f** of the body housing **210** in a complementary recess. When the valve cap **612** is properly seated in the second cavity **210f**, fluid is able to flow therethrough. When the valve cap **612** is not properly seated in the second cavity **210f**, the valve cap **612** acts to seal the fluid within the fluid supply tank **600**.

(48) The valve cap **612** of the fluid supply tank **600** is removably coupled to the lower tier **614a** via a threaded connection, although other connection types are possible such as a hinged lid, etc. When the fluid supply tank **600** is not removably coupled with the body housing **210**, the valve cap **612** is accessible to a user. The valve cap **612** is configured to be removed from a remainder of the fluid supply tank **600**, e.g., by being unscrewed by hand, to allow fluid to be added into the hollow interior of the fluid supply tank **600**, e.g., by being poured therein through an opening uncovered by removal of the valve cap **612**, and/or for fluid to be removed from the hollow interior of the fluid supply tank **600**, e.g., by being poured out of the opening uncovered by removal of the valve cap **612**.

(49) The lower tier **614b** includes a bleeder valve **616** and a retention depression **618**. As the fluid tank **614** empties of fluid, the bleeder valve **616** is configured to allow for an equalization of pressure in the hollow interior to facilitate a constant supply of fluid to the cleaning device **10**, without creating a vacuum within the hollow interior. The retention depression **618** in this illustrated embodiment is a depression formed in the lower tier **614b** which is shaped to receive the tank engagement feature **216** of the body assembly **200** although other configurations are possible for a retention feature configured to engage the tank engagement feature **216**. As explained above, actuation of the fluid tank switch **212** is configured to allow for the fluid supply tank **600** to be removed from the second cavity **210f**. More specifically, actuation of the fluid tank switch **212** is configured to retract the tank engagement feature **216** into the body housing **210** so that it no longer engages the retention depression **618**.

(50) FIG. **13** illustrates various components, including tubing **620**, a fluid pump **622** (also shown in FIG. **18**), and fluid application face **624**, and spray nozzles **630**, configured to facilitate delivery of fluid from the fluid supply tank **600** to a surface for cleaning. The fluid pump **622** is configured to pump fluid from the fluid supply tank **600** through the cleaning device **10** by providing a force that draws fluid out of the fluid supply tank **600**. The fluid supply tank **600**, when removably coupled with the body assembly **200**, is in fluid communication with the spray nozzles **630** by way of the fluid pump **622** and the tubing **620**. When fluid leaves the fluid supply tank **600**, e.g., under a

pumping force provided by the pump **622**, the fluid is transported through the cleaning device **10** in the tubing **620**. The tubing **620** connects to the fluid supply tank **610**, travels down the body assembly **200**, and then travels into the head assembly **100**. More particularly, the tubing **620** connects the fluid supply tank **600** to the pump **622** and then leaves the pump **622** before splitting and finally connecting to left and right spray nozzles **630** disposed on the fluid application face **624** of the head assembly **100**. In other embodiments, the spray nozzles **630** can be located elsewhere in the head assembly **100** and/or can include a different number of nozzles.

(51) Waste (if any) outside the device **10** mixes with the emitted fluid and creates a slurry which is then suctioned into the cleaning device **10** through a central intake (obscured in the figures) of the head assembly **100**. From the central intake, the slurry travels up through the hosing **230** and into the recovery tank **420**.

(52) The cleaning device **10** is configured to allow a user to select wet and dry cleaning modes using the user interface **302**. The user interface **302** is located at the handle assembly **300** in this illustrated embodiment, as shown in FIGS. **2**, **14** and **17** in which the user interface **302** is located on a handle frame **312** of the handle assembly **300**, but the user interface **302** can also or instead be located elsewhere, such as at another area of the handle assembly **300**, at the head assembly **100**, and/or at the body assembly **200**.

(53) As shown in FIGS. **14** and **17**, the handle **310** of the handle assembly **300** includes the handle frame **312** and the stem **320**. The stem **320** defines and surrounds an interior handle aperture **314**. The stem **320** in this illustrated embodiment is substantially linear and has a rear face **322** and a front face **324**. A person skilled in the art will appreciate that an element may not be precisely linear but nevertheless be considered substantially linear for any number of reasons, such as manufacturing tolerances and sensitivity of measurement equipment. A person skilled in the art will also appreciate that the handle assembly **300** can have a variety of other configurations.

(54) As shown in FIGS. **14-17**, the illustrated handle frame **312** has a bottom section **312a**, a front section **312b**, and a back section **312c** extending upward from the bottom section **312a** at substantially right angles relative to the bottom section **312a**. A person skilled in the art will appreciate that an angle may not be precisely a right angle but nevertheless be considered a substantially right angle for any number of reasons, such as manufacturing tolerances and sensitivity of measurement equipment. A top of each of the front section **312b** and the back section **312c** are connected by a top section **312d** of the handle frame **312**. The user interface **302** is located on the front section **312b** of the handle frame **312** in this illustrated embodiment but may be located elsewhere, such as on another section **312a**, **312c**, **312d** of the handle frame **312**, on the body assembly **200**, and/or on the head assembly **100**.

(55) The user interface **302** can have a variety of configurations. In some embodiments, the user interface **302** includes a touchscreen display configured to receive user inputs by touch and to display information thereon for visualization by a user. In some embodiments, the user interface **302**, whether including a touchscreen display or not, includes a plurality of input buttons and/or other controls that permit a user to configure operation of the cleaning device **10** by providing various inputs as described herein. For example, in some embodiments, the user interface **302** can include a rotary function dial configured to rotate when turned by a user such that the user can select wet and dry cleaning modes mode in which the cleaning device **10** can operate. For another example, the user interface **302** can include a start/stop button, which, when pressed by a user, causes the controller **350** to start and/or stop various operation of the cleaning device **10** in the selected mode. For yet another example, the user interface **302** can include one or more indicators (e.g., light(s), display(s), speaker(s), etc.) which indicate a status of the cleaning device **10**. The one or more indicators can include, for example, any one or more of a fluid supply tank empty indicator that indicates whether the fluid supply tank **600** is empty, a mode indicator that indicates which cleaning mode is selected, a power status indicator that indicates whether the cleaning device **10** is powered on, etc.

(56) The handle assembly **300** also includes a power button **330** disposed on the front section **312b** of the handle frame **312**, although the power button **330** may be located elsewhere, such as on the back section **312c**, the stem **320**, as part of the user interface **302**, or elsewhere. The power button **330** is configured to be actuated by a user to turn the device **10** on and off.

(57) The handle assembly **300** also includes an area rug button **340** disposed on a front exterior of the top section **312d** of the handle frame **312**, although the area rug button **340** may be located elsewhere, such as on the back section **312c**, the stem **320**, as part of the user interface **302**, or elsewhere. The area rug button **340** is configured to be actuated by a user to activate an area rug mode when an area rug is to be cleaned using the cleaning device **10**. In the area rug mode, some bleed air is provided to reduce air flow at the central intake.

(58) In some embodiments, the power button **330** and/or the area rug button **340** can be omitted with the functionality of the omitted button(s) instead being provided via the user interface **302**.

(59) The user interface **302**, the power button **330**, and the area rug button **340** are in operable communication with a controller **350**, which is illustrated in FIG. **18**. The user interface **302**, the power button **330**, and the area rug button **340** are configured to receive inputs from a user that cause the controller **350** to perform one or more operations responsive to the received inputs, as described further herein.

(60) The controller **350** of the cleaning device **10** is configured to be in operable communication with various components of the cleaning device **10**, including the user interface **302**, the power button **330**, the area rug button **340**, the pump **622**, and a power supply **360** (e.g., a battery or other source of power). As in this illustrated embodiment, the controller **350** can include a processor **352** and a memory **354** configured to store instructions which, when executed by the processor **352**, cause the processor **352** to perform operations. The controller **350** in this illustrated embodiment also includes an input/output (I/O) interface **356** that enables the processor **350** to receive commands and/or data from other components of the cleaning device **10** for use in performing the operations. For example, the controller **350** can receive, through the I/O interface **356**, data, e.g., a voltage value, from the fluid pump **622m** characterizing an electromotive force (EMF) and provide that data to the processor **352** for use in performing operations requiring the received EMF data as an input, as described further herein. Similarly, the controller **350** can receive, from the user interface **302**, the power button **330**, and the area rug button **340**) and via the I/O interface **68**, data characterizing inputs received from the user by the user interface **302**, the power button **330**, and the area rug button **340**) and provide that data to the processor **352** for use in performing operations that require the data received therefrom as input.

(61) As shown in FIG. **18**, the cleaning device **10** includes a printed circuit board (PCB) **370** including various components, which in this illustrated embodiment include the controller **350**, configured to facilitate operation of the cleaning device **10**. The PCB **370** can have a variety of configurations and, in some embodiments, the controller **350**) can be included in the cleaning device **10** without use of a PCB.

(62) The PCB **370** is located at a control unit **380** of the cleaning device **10**. The control unit **380** is located at the handle assembly **300** in this illustrated embodiment, as shown in FIG. **16**, but can be located elsewhere, such as at the head assembly **100** or the body assembly **200**.

(63) The cleaning device **10** also includes the power supply **360** that is configured to supply power to various components of the cleaning device **10** requiring power to operate. The power supply **360** is in operable communication with the controller **350**. The power supply **360** is configured to receive commands from the controller **350**, provided via the I/O interface **356**, that cause the power supply **360** to provide electrical power to components as needed in reply to the power button **330** being actuated and to cease providing electrical power to components as needed in reply to the power button **330** being actuated again.

(64) The fluid pump **622** is an electric pump that includes a pump motor **622m** configured to provide a driving force that causes fluid to be drawn out of the fluid supply tank **600**. In an

exemplary embodiment, the motor **622m** is a brushless direct current (BLDC) motor or other type of motor that includes a rotor **622r** and a stator **622s**. The rotor **622r** is configured to rotate. The rotor **622r** can include a permanent magnet, and the stator **622s** can include metallic coils electrically coupled to a DC source **638** (see FIG. **19**). DC current provided to the stator **622s** from the DC source **638** is configured to create an electromagnetic field, which is configured to cause the rotor **622r** to rotate such that the pump motor **622m** can provide a rotational driving force configured to cause fluid to be drawn out of the fluid supply tank **600**. The fluid pump **622** is operably coupled with the controller **350**, as shown in FIG. **18**, to facilitate fluid supply tank **600** empty detection based on the created EMF, as discussed further herein.

(65) One exemplary embodiment of a control circuit **619** including the fluid pump **622** is shown in FIG. **19** and includes a MOSFET **632**, a first transistor **634**, a second transistor **636**, and the DC source **638**. The DC source **638** is a 14 V source in this illustrated embodiment, although other voltage values are possible.

(66) During a wet cleaning operation, the amount of fluid in the fluid supply tank **600** decreases because fluid is pumped out of the fluid supply tank **600**. In the course of performing one or more wet cleaning operations, the fluid supply tank **600** will become substantially depleted of fluid and thus be substantially empty. A person skilled in the art will appreciate that a fluid supply tank may not be completely empty, e.g., because one or more droplets of fluid are stuck to an interior surface of the fluid supply tank, but nevertheless be considered to be substantially empty.

(67) When the supply of fluid in the fluid supply tank **600** is determined to be empty, as discussed further herein, the cleaning device **10** is prevented from operating in a wet cleaning operation, e.g., by the controller **350**) causing the pump **622** to stop pumping fluid (or not start if the pump **622** was not already pumping fluid). Additionally, the cleaning device **10** is configured to provide a user notification, e.g., the controller **350** is configured to cause the user notification to be provided via the user interface **302**, that the fluid supply tank **600** must be refilled (or replaced) before a wet cleaning operation can begin, or, if a wet cleaning operation was in progress, can continue. The user notification can be visual and/or audible. The user notification can include, for example, a water droplet or other symbol being shown on a display of the user interface **302**. The user notification can include, for example, text being shown on a display of the user interface **302**. For another example, the user notification can include a light illuminating (solid illumination or blinking illumination) via the user interface **302**. For yet another example, the user notification can include an audible one or more beeps or other sounds provided via the user interface **302**.

(68) In an exemplary embodiment, the controller **350** (e.g., the processor **352** thereof) is configured to determine whether the fluid supply tank **600** is empty using EMF data received from the pump **622**. The EMF data is indicative of the EMF created by the pump motor **622m** and, more particularly, the EMF generated by the motor as a rotating element of the motor **622m**, e.g., the rotor **622r** of the pump motor **622m**, rotates while the pump **622** is running in a pulse width modulated duty cycle.

(69) In response to the fluid pump **622** being turned on and receiving a control signal, e.g., a pulse width modulated (PWM) signal, from the first transistor **634**, the MOSFET **632** receives DC current from the DC source **638**. The pump motor's rotor **622r** will therefore begin to rotate and an EMF will be created. The MOSFET **632** is generally acting as a generator. The controller **350**) is configured to receive a signal from the fluid pump **622** indicative of the EMF. The EMF runs in the circuit **619** to the controller **350**, and in particular to the second transistor **636** of the controller **350**).

(70) In response to the fluid pump **622** being turned off, the MOSFET **632** stops receiving DC current from the DC source **638** and the rotor **622r** thus stops rotating. The rotor's rotation does not stop immediately when the fluid pump **622** is turned off. Instead, inertia causes the rotor **622r** to continue rotating for a period of time after the fluid pump **622** has been turned off and stops receiving DC current. The EMF will continue being created during this period of time. The

controller **350**) (e.g., the second transistor **636** thereof) continues receiving the EMF signal for the period of time after the fluid pump **622** has been turned off.

(71) The controller **350** (e.g., the processor **352** thereof) is configured to determine whether the fluid supply tank **600** is empty based on the received EMF signal. In general, if the EMF signal is less than a predetermined threshold EMF value, then the fluid supply tank **600** is determined to not be substantially empty, and if the EMF signal is greater than the predetermined threshold EMF value, then the fluid supply tank **600** is determined to be substantially empty. As discussed herein, in response to determining that the fluid supply tank **600** is substantially empty, the controller **350**) (e.g., the processor **352** thereof) is configured to cause a user notification to be provided indicating that the fluid supply tank **600** must be refilled (or replaced).

(72) With fluid present in the fluid supply tank **600**, the fluid pump **622** runs with a load. Conversely, with the fluid supply tank **600** being substantially empty, the load is not present. The EMF created after the pump **622** has been turned off while the rotor **622r** is still rotating due to inertia will therefore be higher when the fluid supply tank **600** is substantially empty than when the fluid supply tank **600** is not substantially empty.

(73) FIG. **20** illustrates a graph **700** showing one embodiment of EMF versus time. FIG. **20** shows two periods of time **702**, **704** when a fluid supply tank (e.g., fluid supply tank **600** or other fluid supply tank) is substantially empty and another period of time **706** when the fluid supply tank is not substantially empty. As shown in FIG. **20**, the EMF is higher when the fluid supply tank is substantially empty. The EMF can therefore be used by the controller **350** is determining whether the fluid supply tank **600** is substantially empty or not.

(74) The predetermined threshold EMF value is preset. e.g., stored in the memory **354** of the controller **350**, for a particular cleaning device **10**. In an exemplary embodiment, the predetermined threshold EMF value is about halfway between an EMF value with the fluid supply tank **600** being substantially empty and an EMF value with the fluid supply tank **600** not being substantially empty. Being about halfway between these EMF values may help prevent false positives, e.g., prevent the controller **350** from determining that the fluid supply tank **600** is substantially empty when the fluid supply tank **600** still contains fluid therein that could be pumped out of the fluid supply tank **600** and delivered from the cleaning device **10** to a surface. Different cleaning devices can have different fluid supply tanks and/or different fluid pumps, so the predetermined threshold EMF value can vary for different cleaning devices. The predetermined threshold EMF value can thus be determined experimentally. In the example of FIG. **20**, the predetermined threshold EMF value can be selected to be a voltage value of about 30,000, which is about halfway between an EMF value with the fluid supply tank **600** is substantially empty and an EMF value with the fluid supply tank **600** is not substantially empty.

(75) The predetermined threshold value EMF is valid even if the cleaning device **10** is operating in different wet cleaning modes. In some embodiments, the cleaning device **10** is configured to operate in different wet cleaning modes, such as a bare floor wet cleaning mode, rug wet cleaning mode, and cleaning device self-cleaning mode. In different wet cleaning modes, the controller (e.g., the processor **352**) thereof is configured to cause fluid to be pumped from the fluid supply tank **600** at different rates. For example, fluid can be pumped from the fluid supply tank **600** at a lower rate for cleaning bare floors than for cleaning a rug. For another example, fluid can be pumped from the fluid supply tank **600** at a higher rate for cleaning device **10** self-cleaning than for surface cleaning.

(76) FIG. **21** illustrates a graph **800** showing one embodiment of EMF versus time for a bare floor wet cleaning mode. FIG. **22** illustrates a graph **802** showing one embodiment of EMF versus time for a rug wet cleaning mode. FIG. **23** illustrates a graph **804** showing one embodiment of EMF versus time for a first, low flow self-cleaning cleaning mode, and FIG. **24** illustrates a graph **806** showing one embodiment of EMF versus time for a second, high flow self-cleaning cleaning mode. As demonstrated by FIGS. **21-24**, a predetermined threshold EMF value of about 30,000 is valid for all of the illustrated wet cleaning modes.

(77) FIG. 25 shows first and second graphs **900**, **902** over time and illustrating EMF being higher when a fluid supply tank of a cleaning device (e.g., the cleaning device **10** or other cleaning device) is substantially empty as compared to when the fluid supply tank is not substantially empty. The first graph **900** shows a first line **904** representing power to a fluid pump of the cleaning device (e.g., the fluid pump **622** or other fluid supply tank) and a second line **906** representing an EMF created at the fluid pump, e.g., by rotation of a rotor of the fluid pump. The EMF begins to change when the power to the fluid pump stops and begins to decrease, as shown by the second line **906** sloping downward at a first slope after a first period of time has elapsed from the stop of power. The second graph **902** shows a third line **908** representing power to the fluid pump and a fourth line **910** representing an EMF created at the fluid pump. The EMF begins to change when the power to the fluid pump stops and begins to decrease, as shown by the fourth line **910** sloping downward at a second slope after a second period of time has elapsed from the stop of power.

(78) As shown in FIG. 25, the first period of time is less than the second period of time, which reflects that the load experienced by the fluid pump is greater in the scenario shown in the first graph **900** than in the scenario shown in the second graph **902** because the fluid supply tank is not substantially empty in the scenario illustrated in the first graph **900** and is substantially empty in the scenario illustrated in the second graph **902**. The first line **904** therefore begins sloping downward sooner in response to power stopping than the third line **908** begins sloping downward in response to power stopping because the fluid pump's rotor stops rotating sooner in the scenario illustrated in the first graph **900** than in the scenario illustrated in the second graph **902**. Also, the first slope of the first line **904** is greater than the second slope of the third line **908**, which also reflects that the load experienced by the fluid pump is greater in the scenario shown in the first graph **900** than in the scenario shown in the second graph **902**.

(79) FIG. 26 illustrates one embodiment of a method **1000** of fluid tank empty detection. The method **1000** is described with respect to the cleaning device **10** of FIGS. 1-4 for ease of explanation but can be similarly performed with respect to another cleaning device or another type of device having a fluid supply tank configured to be refilled (or replaced).

(80) The method **1000** includes the fluid pump **622** pumping **1002** fluid from the fluid supply tank **600**. The pumping can start pumping **1002** by a user providing an input to the cleaning device **10**, e.g., by the user pressing an on/off switch (as shown in FIG. 26) or by the user providing another input. The user's input is received by the controller **350** (e.g., the processor **352** thereof) and causes the controller **350** (e.g., the processor **352** thereof) to transmit a signal to the fluid pump **622** that causes the fluid pump **622** to start pumping and thereby start drawing fluid out of the fluid supply tank **600**.

(81) After running for a period of time during which fluid is drawn from the fluid supply tank **600**, the fluid pump **622** stops **1004** pumping. The pump **622** is shown in FIG. 26 as pumping **1002** for five seconds with the pump **622** running at 5 Hz before stopping **1004**, but another period of time of pumping **1002** and another frequency are possible. The pumping can stop **1004** pumping by a user providing an input to the cleaning device **10**, e.g., by the user pressing an on/off switch or by the user providing another input. The user's input is received by the controller **350** (e.g., the processor **352** thereof) and causes the controller **350** (e.g., the processor **352** thereof) to transmit a signal to the fluid pump **622** that causes the fluid pump **622** to stop **1004** pumping and thereby stop drawing fluid out of the fluid supply tank **600**.

(82) The controller **350** (e.g., the processor **352** thereof) waits **1004** for a predetermined period of time delay after the pumping stops **1004** before the controller **350** (e.g., the processor **352** thereof) begins reading **1004** EMF data (identified as "Back_EMF value" in FIG. 26) from the fluid pump **622** on a regular periodic basis. The regular periodic basis is every 2 ms in FIG. 26 but other regular periodic bases are possible. The predetermined period of time delay is 5 ms in FIG. 26, but another predetermined period of time delay is possible. Waiting the predetermined period of time delay, e.g., as counted by a counter or timer of the controller **350**) that is in operable

communication with the processor **352**, reflects noise at the pump **622** immediately after stopping **1004**. The fluid pump **622** continues running for a period of time after stopping **1004** because the electric motor **622m** of the fluid pump **622** cannot instantaneously stop completely, as discussed above. The first and third lines **904**, **908** of the first and second graphs **900**, **902**, respectively, of FIG. **26** reflect noise in such a period of time before the first and third lines **904**, **908** begin sloping downward.

(83) After each read **1004** of the EMF data, the controller **350** (e.g., the processor **352** thereof) determines **1006** whether a predetermined number of EMF data reads **1004** have occurred, e.g., if a predetermined number of EMF data points have been acquired. The predetermined number of EMF data reads is twenty in FIG. **26**, but other numbers are possible. If the predetermined number of EMF data reads **1004** has not occurred, then the controller **350** (e.g., the processor **352** thereof) continues reading **1004** EMF data.

(84) If the predetermined number of EMF data reads **1004** has occurred, then the controller **350** (e.g., the processor **352** thereof) calculates **1008** a sum of the predetermined number of EMF data reads **1004**, e.g., adds together the acquired EMF values. The calculation **1008** roughly calculates an area of the back electromotive force of the fluid pump **622**. For example, with reference to the example of FIG. **25**, the calculation **1008** for EMF data of the first graph **900** roughly calculates an area under the first line **904** and for EMF data of the second graph **902** roughly calculates an area under the third line **908**.

(85) Referring again to FIG. **26**, the controller **350** (e.g., the processor **352** thereof) then determines **1010** whether the sum continuously exceeds the predetermined threshold EMF value (identified as “threshold A” in FIG. **26**) for a predetermined amount of time. The predetermined amount of time is three seconds in FIG. **26**, but another predetermined amount of time is possible. If the sum does not continuously exceed the predetermined threshold EMF value for the predetermined amount of time, the controller **350** (e.g., the processor **352** thereof) continues reading **1004** EMF data from the fluid pump **622** on the regular periodic basis. This reflects that the fluid supply tank **600** is not substantially empty and that fluid can therefore continue being pumped from the fluid supply tank **600**.

(86) If the sum does continuously exceed the predetermined threshold EMF value for the predetermined amount of time, the controller **350** (e.g., the processor **352** thereof), the fluid pump **622** stops **1012** running and a user notification is provided **1012** that the fluid supply tank **600** must be refilled (or replaced) before a wet cleaning operation can begin, or, if a wet cleaning operation was in progress, can continue. This reflects that the fluid supply tank **600** is substantially empty and that fluid cannot be pumped from the fluid supply tank **600** until the fluid supply tank **600** is refilled (or replaced). e.g., that the method **1000** has ended **1014** until the fluid supply tank **600** is refilled (or replaced) and pumping **1002** may begin again. The fluid pump **622** can stop **1012** running by the controller **350** (e.g., the processor **352** thereof) transmitting a signal to the fluid pump **622** that causes the pump **622** to stop pumping, e.g., for the motor **622m** to stop running. The user notification can be provided **1012** in any number of ways, as discussed above.

(87) FIG. **27** illustrates another embodiment of a method **1100** of fluid tank empty detection. The method **1100** is described with respect to the cleaning device **10** of FIGS. **1-4** for ease of explanation but can be similarly performed with respect to another cleaning device or another type of device having a fluid supply tank configured to be refilled (or replaced).

(88) In the method **1000** of FIG. **26**, the predetermined threshold EMF value is preset and remains the same throughout performance of the method **1000**. Conversely, in the method **1100** of FIG. **27**, the predetermined threshold EMF value is initially at a preset value, but that value may change during performance of the method **1100**. The method **1100** of FIG. **27** reflects that the cleaning device **10** can be configured to perform self-learning or machine learning that results in the predetermined threshold EMF value being changed to another value. Self-learning or machine learning may allow for more accurate empty tank detection over time because electric motors

experience wear and tear over time that can affect the created EMF. As discussed further below, the self-learning or machine learning is only conducted when the pump **622** is running and the EMF value is less than the threshold value which means that the fluid supply tank **600** is not empty.

(89) The method **1100** includes the fluid pump **622** pumping **1102** fluid from the fluid supply tank **600**. The pumping can start pumping **1102** by a user providing an input to the cleaning device **10**, e.g., by the user pressing an on/off switch (as shown in FIG. **27**) or by the user providing another input. The user's input is received by the controller **350** (e.g., the processor **352** thereof) and causes the controller **350** (e.g., the processor **352** thereof) to transmit a signal to the fluid pump **622** that causes the fluid pump **622** to start pumping and thereby start drawing fluid out of the fluid supply tank **600**. The user's input being received by the controller **350** (e.g., the processor **352** thereof) also causes the controller **350** (e.g., the processor **352** thereof) to read **1102** the predetermined threshold EMF value (identified as "flash data threshold A" in FIG. **27**) from the memory **354**.

(90) After running **1104** for a period of time during which fluid is drawn from the fluid supply tank **600**, the controller **350** (e.g., the processor **352** thereof) begins reading **1106** EMF data (identified as "Back_EMF Data" in FIG. **27**) from the fluid pump **622** on a regular periodic basis. The pump **622** is shown in FIG. **27** as running **1104** for five seconds, but another period of time of running **1104** is possible. The regular periodic basis is every 1 second in FIG. **27** but other regular periodic bases are possible. After each read **1106** of the EMF data, the controller **350** (e.g., the processor **352** thereof) determines **1108** whether a predetermined number of EMF data reads **1106** have occurred, e.g., if a predetermined number of EMF data points have been acquired. The predetermined number of EMF data reads is thirty-two in FIG. **27**, but other numbers are possible.

(91) If the predetermined number of EMF data reads **1106** has occurred, then the controller **350** (e.g., the processor **352** thereof) calculates **1110** a potential new predetermined threshold EMF value (identified as "threshold F for self-learning" in FIG. **27**). The potential new predetermined threshold EMF value is calculated **1110** by averaging the read **1106** EMF data and adding a predetermined value. For example, for an initial predetermined threshold EMF value of 30,000 (as discussed above) the predetermined value added to the average EMF can be 20,000. The potential new predetermined threshold EMF value can be calculated **1110** by the controller **350** (e.g., the processor **352** thereof) only when the fluid supply tank **600** had at least some fluid therein that the pump **622** pumped from the fluid supply tank **600**. Otherwise, no EMF data could have been acquired **1106** for performance of the calculation **1110**.

(92) After calculating **1110** the potential new predetermined threshold EMF value, the controller **350** (e.g., the processor **352** thereof) determines **1112** whether the potential new predetermined threshold EMF value is within a predetermined range of the predetermined threshold EMF value. The predetermined range is $\pm 5,000$ in FIG. **27**, but another predetermined range is possible. If potential new predetermined threshold EMF value is within the predetermined range of the predetermined threshold EMF value, the controller **350** (e.g., the processor **352** thereof) resets **1114** the calculator (e.g., the controller **350** (e.g., the processor **352** thereof) that calculated **1110** the potential new predetermined threshold EMF value resets) and continues reading **1106** EMF data from the fluid pump **622** on the regular periodic basis. This reflects that the predetermined threshold EMF value remains the same.

(93) If the potential new predetermined threshold EMF value is not within the predetermined range of the predetermined threshold EMF value, the potential new predetermined threshold EMF value is assigned **1116** as the predetermined threshold EMF value. Such assigning **1116** can be performed by the controller **350** (e.g., the processor **352** thereof) causing the potential new predetermined threshold EMF value to be stored as the predetermined threshold EMF value in volatile memory of the memory **624**.

(94) After setting **1116** the new predetermined threshold EMF value, the controller **350** (e.g., the processor **352** thereof) determines **1118** whether the fluid pump **622** has not stopped running, which is shown in FIG. **27** as the cleaning device **10** entering a standby mode. If the fluid pump **622** has

stopped running, then the controller **350** (e.g., the processor **352** thereof) continues reading **1106** EMF data.

(95) If the fluid pump **622** has stopped running **1004**, then the controller **350** (e.g., the processor **352** thereof) saves **1120** the potential new predetermined threshold EMF value as the predetermined threshold EMF value. Such saving **1120** can be performed by the controller **350** (e.g., the processor **352** thereof) causing the potential new predetermined threshold EMF value to overwrite the predetermined threshold EMF value in non-volatile memory (shown in FIG. **27** as “flash” memory) of the memory **624**. The method **1100** then ends **1122** until the pumping begins again.

(96) If the predetermined number of EMF data reads **1106** has not occurred, then the controller **350** (e.g., the processor **352** thereof) determines **1124** whether the pump **622** has stopped running. If not, then the controller **350** (e.g., the processor **352** thereof) continues reading **1106** EMF data. If so, then the controller **350** (e.g., the processor **352** thereof) resets **1126** the calculator, similar to the resetting **1114** discussed above, and the method ends **1122** as discussed above.

(97) The methods **1000**, **1100** of FIGS. **26** and **27** can each be implemented in the cleaning device **10** such that the device **10** is configured to perform self-learning. Alternatively, only the method **1000** of FIG. **26** can be implemented in the cleaning device **10** such that the device **10** is not configured to perform self-learning.

(98) The subject matter described herein can be implemented in analog electronic circuitry, digital electronic circuitry, and/or in computer software, firmware, or hardware, including the structural means disclosed in this specification and structural equivalents thereof or in combinations of them. The subject matter described herein can be implemented as one or more computer program products, such as one or more computer programs tangibly embodied in an information carrier (e.g., in a machine-readable storage device), or embodied in a propagated signal, for execution by, or to control the operation of, data processing apparatus (e.g., a programmable processor, a computer, or multiple computers). A computer program (also known as a program, algorithm, software, software application, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file. A program can be stored in a portion of a file that holds other programs or data, in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub-programs, or portions of code).

(99) The processes and logic flows described in this specification, including the method steps of the subject matter described herein, can be performed by one or more programmable processors executing one or more computer programs to perform functions of the subject matter described herein by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus of the subject matter described herein can be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit).

(100) Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processor of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for executing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. Information carriers suitable for embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, (e.g., EPROM, EEPROM, and flash memory devices). The processor and the

memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(101) The techniques described herein can be implemented using one or more modules. As used herein, the term “module” refers to computing software, firmware, hardware, and/or various combinations thereof. At a minimum, however, modules are not to be interpreted as software that is not implemented on hardware, firmware, or recorded on a non-transitory processor-readable recordable storage medium (i.e., modules are not software per se). Indeed “module” is to be interpreted to always include at least some physical, non-transitory hardware such as a part of a processor or computer. Two different modules can share the same physical hardware (e.g., two different modules can use the same processor). The modules described herein can be combined, integrated, separated, and/or duplicated to support various applications. Also, a function described herein as being performed at a particular module can be performed at one or more other modules and/or by one or more other devices instead of or in addition to the function performed at the particular module.

(102) One skilled in the art will appreciate further features and advantages of the devices, systems, and methods based on the above-described embodiments. Accordingly, this disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated herein by reference in their entirety for all purposes.

(103) The present disclosure has been described above by way of example only within the context of the overall disclosure provided herein. It will be appreciated that modifications within the spirit and scope of the claims may be made without departing from the overall scope of the present disclosure.

Claims

1. A system comprising: a fluid pump of a cleaning device configured to pump a fluid from a fluid supply tank for delivery to a surface to be cleaned by the cleaning device, the fluid pump including a motor configured to drive the pumping of the fluid and to generate an electromotive force (EMF); and a controller configured to receive from the fluid pump a signal indicative of the EMF and configured to determine, based on the received signal, whether the fluid supply tank is substantially empty; wherein determining the fluid supply tank is substantially empty includes comparing EMF data from after the fluid pump has been turned off with a pre determined threshold EMF value.
2. The system of claim 1, wherein the controller is configured to, in response to determining that the fluid supply tank is substantially empty, cause a user notification to be provided via the cleaning device indicating that the fluid supply tank is substantially empty.
3. The system of claim 1, wherein the controller receiving the signal includes the controller receiving a plurality of signals from the fluid pump, each of the plurality of signals being indicative of the EMF generated by the motor during a period of time; and the controller determining whether the fluid supply tank is substantially empty includes the controller comparing the plurality of signals with the predetermined threshold EMF value.
4. The system of claim 3, wherein the predetermined threshold EMF value is a preset value that does not change.
5. The system of claim 1, further comprising the fluid supply tank.
6. The system of claim 1, wherein the motor includes a rotor and a stator.
7. The system of claim 1, further comprising a user interface configured to receive a user instruction that causes the fluid pump to be turned off.
8. The system of claim 3, wherein the controller is configured to determine whether the plurality of signals are outside of a predetermined range; if the plurality of signals are not outside of the predetermined range, the predetermined threshold EMF value remains the same; and if the plurality of signals are outside of the predetermined range, the controller is configured to change the

predetermined threshold EMF value to a new predetermined threshold EMF value.

9. The system of claim 8, wherein the controller is configured to calculate the new predetermined threshold EMF value based on the plurality of signals.

10. A system, comprising: a fluid pump of a cleaning device configured to pump a fluid from a fluid supply tank for delivery to a surface to be cleaned by the cleaning device, the fluid pump including a motor configured to drive the pumping of the fluid and to generate an electromotive force (EMF); and a controller configured to receive from the fluid pump a signal indicative of the EMF and configured to determine, based on the received signal, whether the fluid supply tank is substantially empty; wherein the controller receiving the signal includes the controller receiving a plurality of signals from the fluid pump, each of the plurality of signals being indicative of the EMF generated by the motor during a period of time; and the controller determining whether the fluid supply tank is substantially empty includes the controller comparing the plurality of signals with a predetermined threshold EMF value; and the controller is configured to determine whether the plurality of signals are outside of a predetermined range; if the plurality of signals are not outside of the predetermined range, the predetermined threshold EMF value remains the same; and if the plurality of signals are outside of the predetermined range, the controller is configured to change the predetermined threshold EMF value to a new predetermined threshold EMF value.

11. The system of claim 10, wherein the controller is configured to calculate the new predetermined threshold EMF value based on the plurality of signals.

12. A system comprising: a processor; and a memory storing instructions that, when executed by the processor, cause the processor to perform operations comprising: causing a fluid pump of a cleaning device to pump a fluid from a fluid supply tank for delivery to a surface to be cleaned by the cleaning device, the fluid pump including a motor configured to drive the pumping of the fluid and to generate an electromotive force (EMF), and determining, based on the EMF generated after the fluid pump has been turned off, whether the fluid supply tank is substantially empty.

13. The system of claim 12, wherein the operations further comprise, in response to determining that the fluid supply tank is substantially empty, causing a user notification to be provided via the cleaning device indicating that the fluid supply tank is substantially empty.

14. The system of claim 12, wherein determining whether the fluid supply tank is substantially empty includes comparing EMF data received by the processor from the fluid pump with a predetermined threshold EMF value.

15. The system of claim 14, wherein the predetermined threshold EMF value is a preset value that does not change.

16. The system of claim 14, wherein the operations further comprise determining whether the EMF data are outside of a predetermined range; if the EMF data is not outside of the predetermined range, the predetermined threshold EMF value remains the same; and if the EMF data is outside of the predetermined range, the operations further comprise changing the predetermined threshold EMF value to a new predetermined threshold EMF value.

17. The system of claim 16, wherein the operations further comprise calculating the new predetermined threshold EMF value based on the EMF data.

18. The system of claim 12, further comprising the fluid supply tank.

19. The system of claim 12, wherein the operations further comprise causing the fluid pump to be turned off in response to a user input to the cleaning device.
